

Full-Stack-Job-Board — Cheat Sheet

Chain C · Project C-1 · dense one-page recap — pairs with docs/interactive/index.html

1 · Architecture

Layer	Role
Browser	UI state only
FastAPI	Stateless — no session memory
Postgres	Data state (source of truth)

- Every request carries its own JWT — any API replica can answer it
- `Depends(get_db)` injects a scoped DB session per request

2 · JWT Auth

Step	Function
register	<code>hash_password()</code> (bcrypt)
login	<code>verify_password()</code> → <code>create_access_token()</code>
request	<code>Authorization: Bearer <jwt></code>
protect	<code>decode_token()</code> → load user, 401 if invalid

- **Signed ≠ encrypted.** Payload is plain base64 — never put secrets in it
- Signature only proves the token wasn't forged without `SECRET_KEY`

Reference: this repo's domain model

Model	Key fields	Relationship
User	email, hashed_password, is_employer	1:N → Job, 1:N → Application
Job	title, description, location, owner_id	1:N → Application, cascade delete-orphan
Application	job_id, user_id, cover_letter	belongs to Job + User

3 · React Query

Hook	Purpose
<code>useQuery</code>	Fetch + cache under a <code>queryKey</code>
<code>useMutation</code>	POST/PATCH/DELETE
<code>invalidateQueries</code>	Marks a key stale → triggers refetch

- A successful mutation does NOT auto-update other components' cache
- Always pair `onSuccess` with `invalidateQueries({queryKey: [...]})`

4 · Migrations

Command	Effect
<code>alembic revision --autogenerate</code>	Diff models vs. live schema → draft migration
<code>alembic upgrade head</code>	Apply all pending migrations, forward
<code>alembic downgrade -1</code>	Reverse the last migration

- **Autogenerate can't detect renames** — it writes drop + add, which loses data
- Always read a generated migration before running it